

Representation Learning with the Iris Dataset

1 What Will You Learn?

Imagine that you are shown an iris flower and asked:

“Is this flower *Iris setosa*, *Iris versicolor*, or *Iris virginica*?”

A computer cannot look at a flower exactly the way a human does. Instead, we give it measurements.

In this tutorial, we use the famous **Iris dataset** to understand an important idea in artificial intelligence called **representation learning**.

By the end, you should understand:

- how a flower can be represented using numbers,
- how a neural network transforms those numbers,
- what learned weights do,
- what a learned representation or embedding is,
- why similar flowers can become close together in a learned space,
- and how the final representation helps a model classify a flower.

2 The Iris Dataset

The Iris dataset contains measurements from **150 iris flowers** belonging to three species.

Species	Number of Flowers
Iris setosa	50
Iris versicolor	50
Iris virginica	50

For each flower, four measurements are recorded.

Feature	Meaning
Sepal length	Length of the sepal
Sepal width	Width of the sepal
Petal length	Length of the petal
Petal width	Width of the petal

All measurements are in centimeters.

For example, one flower might be represented as

$$x = [5.1, 3.5, 1.4, 0.2].$$

The four numbers mean

$$x = [\text{sepal length}, \text{sepal width}, \text{petal length}, \text{petal width}].$$

The computer does not initially see

“a purple flower with narrow petals.”

Instead, it sees

$$[5.1, 3.5, 1.4, 0.2].$$

This numerical description is called a **representation**.

3 What Is a Representation?

A representation is simply a way of describing something.

A person could be represented by

[height, weight, age].

A song could be represented by

[tempo, volume, pitch].

An iris flower can be represented by

[sepal length, sepal width, petal length, petal width].

These are called **raw features** because humans decided which measurements to collect.

The original Iris representation therefore has four dimensions.

Mathematically,

$$x \in \mathbb{R}^4.$$

Each flower is a point in a four-dimensional feature space.

4 Why Do We Need Representation Learning?

Suppose we have two flowers:

$$x_A = [5.0, 3.4, 1.5, 0.2]$$

and

$$x_B = [5.1, 3.5, 1.4, 0.2].$$

They are very similar.

Now compare them with

$$x_C = [6.7, 3.0, 5.2, 2.3].$$

This flower is quite different.

A good representation should make useful similarities and differences easy to recognize.

That is the goal of **representation learning**.

Instead of relying only on the original measurements, we allow the model to learn a new description:

$$x \rightarrow z,$$

where

x = original features

and

z = learned representation.

5 Representation Learning as Reorganizing Information

Imagine a classroom in which students are initially standing randomly.

Then the teacher says:

“Move closer to students who enjoy similar activities.”

Groups might appear like this:

Music students

Sports students

o o o	^ ^
o o o o	^ ^ ^

Art students

#
#

The students have not changed.

Only their **arrangement** has changed.

Representation learning does something similar.

Conceptually:

Raw measurement space

o	^
#	
	o
^	#
o	

neural network

|

v

Learned representation

Setosa	Versicolor	Virginica
o o o	# #	^ ^
o o	# # #	^ ^ ^

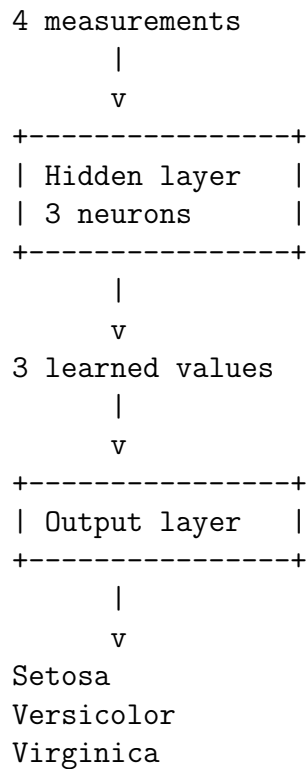
The learned space can make classification easier.

6 A Simple Neural Network

Suppose an input flower is

$$x = \begin{bmatrix} 5.9 \\ 3.0 \\ 4.2 \\ 1.5 \end{bmatrix}.$$

A simple neural network might look like



The hidden layer creates a new representation.

For example,

$$[5.9, 3.0, 4.2, 1.5]$$

might become

$$[1.3, -0.5, 2.1].$$

The first vector represents physical measurements.

The second vector is learned by the neural network.

7 Where Do the New Numbers Come From?

The network calculates them using learned weights.

Suppose one hidden neuron computes

$$z_1 = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b.$$

Here,

$$x_1 = \text{sepal length},$$

$$x_2 = \text{sepal width},$$

$$x_3 = \text{petal length},$$

and

$$x_4 = \text{petal width}.$$

Suppose the model learns

$$w = [0.1, -0.2, 0.8, 1.2].$$

Then

$$z_1 = 0.1(\text{sepal length}) - 0.2(\text{sepal width}) + 0.8(\text{petal length}) + 1.2(\text{petal width}) + b.$$

Petal length and petal width have larger weights.

This means they have a stronger influence on this hidden feature.

8 What Does a Weight Mean?

A weight tells the network how strongly one input should affect a calculation.

For example,

$$z = 0.1x_1 + 0.1x_2 + 2.0x_3 + 3.0x_4.$$

The coefficients of x_3 and x_4 are much larger.

Therefore, petal length and petal width have more influence.

Conceptually:

```
Sepal length --- 0.1 ---\
Sepal width  --- 0.1 ----\
Petal length --- 2.0 -----+---> hidden feature
Petal width  --- 3.0 ----/
```

Before training, weights are usually initialized to small, approximately random values.

During training, the model changes them.

9 How Does the Network Learn the Weights?

Suppose the model receives

$$x = [5.1, 3.5, 1.4, 0.2]$$

and the correct answer is

Setosa.

The model might predict:

Species	Score
Setosa	0.30
Versicolor	0.50
Virginica	0.20

The prediction is wrong because Versicolor received the highest score.

The model calculates an error, usually called a **loss**.

Then an algorithm such as **gradient descent** changes the weights slightly to reduce that error.

Training repeatedly performs

input $\rightarrow$ prediction $\rightarrow$ compare with correct answer $\rightarrow$ calculate loss $\rightarrow$ adjust weights.

After many examples, the weights become useful for distinguishing species.

10 What Are the Weights Learning?

The weights can capture patterns such as:

“Flowers with very short petals are often Setosa.”

or

“Flowers with larger petals are more likely to be Virginica.”

The computer does not store these rules as English sentences.

Instead, the patterns are encoded mathematically in the weights.

A hidden neuron may respond to something resembling

petal size

or

relationship between sepal and petal dimensions.

These combinations were not explicitly programmed by a human.

The network discovered useful features from data.

That is why the process is called **representation learning**.

11 Raw Features Versus Learned Features

The raw representation might be

$$x = [5.9, 3.0, 4.2, 1.5].$$

A learned representation might be

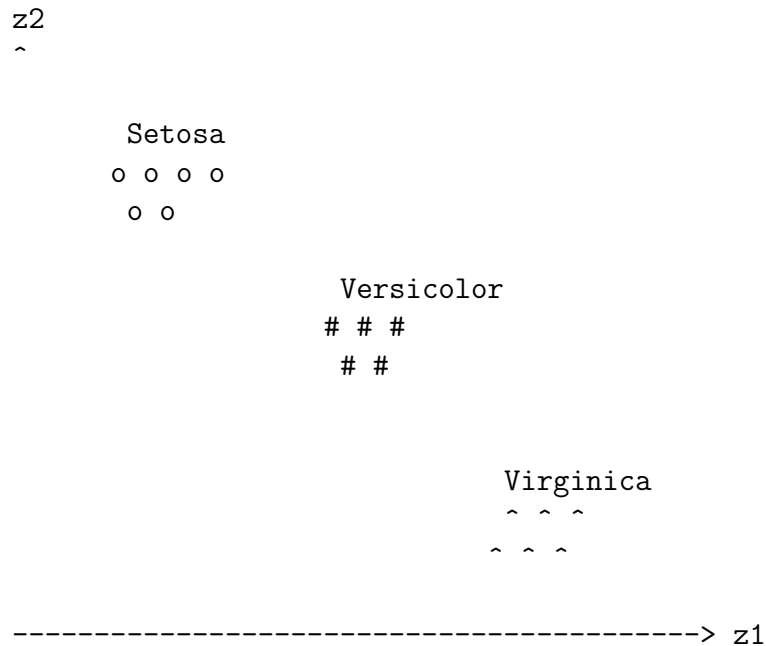
$$z = [2.1, -0.7].$$

The raw values have obvious physical meanings.

The learned values may not have simple human-readable interpretations.

However, they can be very useful for classification.

Conceptually:



12 What Is an Embedding?

A learned representation is often called an **embedding**.

An embedding is a vector of numbers representing an object.

For example,

$$x = [5.9, 3.0, 4.2, 1.5]$$

might become

$$z = [1.4, -0.8].$$

The model has transformed

$$\mathbb{R}^4 \rightarrow \mathbb{R}^2.$$

The goal is to preserve useful information while making the structure of the data easier to use.

13 Why Can Learned Representations Help Classification?

Suppose the model produces these embeddings.

For Setosa:

$$(1.0, 1.1), \quad (1.2, 0.9), \quad (0.8, 1.0).$$

For Versicolor:

$$(4.0, 3.0), \quad (4.2, 3.1), \quad (3.8, 2.9).$$

For Virginica:

$$(7.0, 6.1), \quad (6.8, 5.9), \quad (7.2, 6.0).$$

The species form clusters.

A new point such as

$$z = (4.1, 3.2)$$

is close to the Versicolor cluster.

Representation learning has transformed the problem into a simpler geometric problem.

14 The Final Classification Layer

After obtaining the learned representation, the final layer calculates class scores.

Suppose

$$z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}.$$

For Setosa,

$$s_S = w_S^T z + b_S.$$

For Versicolor,

$$s_V = w_V^T z + b_V.$$

For Virginica,

$$s_G = w_G^T z + b_G.$$

Suppose the scores are

$$s_S = -1.2,$$

$$s_V = 3.4,$$

and

$$s_G = 0.8.$$

The largest score belongs to Versicolor.

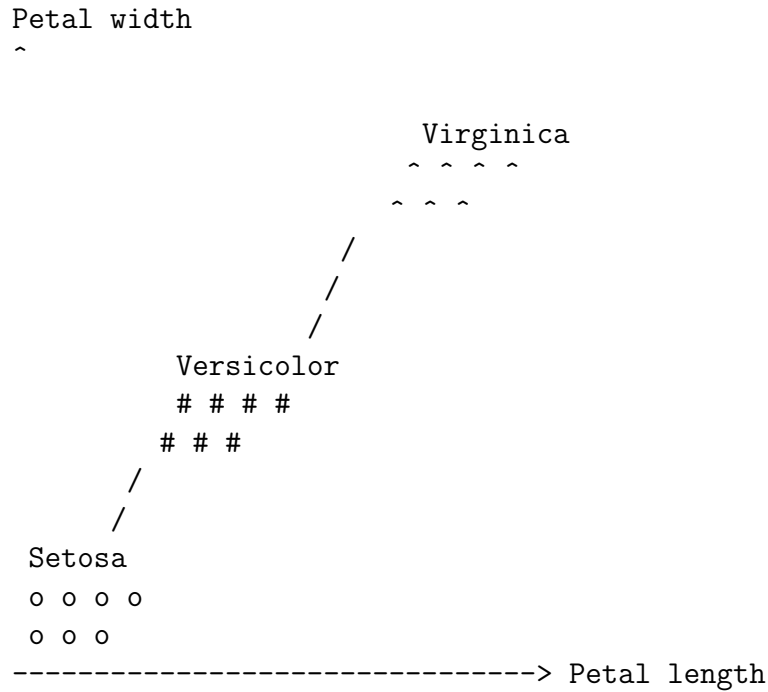
Therefore,

$$\boxed{\text{prediction} = \text{Iris versicolor}}.$$

15 Decision Boundaries

A classifier divides feature space into different regions.

If we use only petal length and petal width, a simplified picture might look like:



The lines represent **decision boundaries**.

Learned weights determine the location and direction of these boundaries.

16 Representation Learning Versus Feature Engineering

Before modern neural networks became popular, people often designed useful features manually.

For example, someone might calculate

$$\text{petal area} \approx \text{petal length} \times \text{petal width}.$$

This is called **feature engineering**.

In feature engineering,

humans design features.

In representation learning,

the model learns features.

Instead of manually deciding which combinations are useful, we allow the model to discover them from training data.

17 A Tiny Neural Network Example

Here is a simple example using Python and PyTorch.

```
import torch
import torch.nn as nn

class IrisNet(nn.Module):
    def __init__(self):
        super().__init__()

        self.hidden = nn.Linear(4, 2)
        self.output = nn.Linear(2, 3)

    def forward(self, x):
        z = torch.relu(self.hidden(x))
        scores = self.output(z)
        return scores, z
```

The network contains

```
4 input features
  |
  v
2 hidden neurons
  |
  v
3 output scores
```

The two hidden neurons create

$$z = [z_1, z_2].$$

This two-dimensional vector is the learned representation.

Because it has only two dimensions, it can be plotted on a graph.

18 Training the Network

Conceptually, training looks like this:

```
for x, correct_species in training_data:

    prediction, representation = model(x)

    loss = compare(prediction, correct_species)

    update_weights_to_reduce(loss)
```

The real implementation uses backpropagation and an optimizer, but the basic logic is

predict $\rightarrow$ measure mistake $\rightarrow$ adjust weights $\rightarrow$ repeat.

19 Before Training Versus After Training

Before training, the hidden-layer weights are close to random.

The representation may look like:

Before training

```
o   ^
   #
 ^   o
   #
   ^
o
```

After training, the classes may become much more organized:

After training

o o o

#

^ ^ ^

A key idea is

Training changes the representation of the data.
--

20 The Model Learns What Matters

Imagine taking photographs of flowers outdoors.

Many things can vary:

- lighting,
- background,
- flower orientation,
- tiny color differences,
- camera angle.

Some differences are useful for identifying species.

Others are irrelevant.

A good representation tries to emphasize

important differences

while reducing the importance of

irrelevant differences.

With the Iris dataset, the model learns which combinations of the four measurements are useful for identifying species.

The same idea applies to larger AI systems.

For image recognition,

pixels $\rightarrow$ edges $\rightarrow$ shapes $\rightarrow$ object representation.

For language models,

text $\rightarrow$ learned embeddings $\rightarrow$ higher-level concepts.

21 Why Do Similar Flowers Become Close Together?

Suppose two Setosa flowers have measurements

$$x_A = [5.1, 3.5, 1.4, 0.2]$$

and

$$x_B = [5.0, 3.4, 1.5, 0.2].$$

The same neural-network transformation is applied to both:

$$z_A = f(x_A)$$

and

$$z_B = f(x_B).$$

Their learned representations might be

$$z_A = [0.9, 1.1]$$

and

$$z_B = [1.0, 1.0].$$

A Virginica flower might be mapped to

$$z_C = [6.8, 5.9].$$

Thus,

$$\text{distance}(z_A, z_B)$$

is small, while

$$\text{distance}(z_A, z_C)$$

is much larger.

This means the representation captures useful structure.

22 Does Every Neural Network Automatically Create Perfect Clusters?

No.

The quality of the learned representation depends on many things:

training data,

model architecture,

loss function,

optimization,

and

amount of training.

A poorly trained model can produce a poor representation.

A model can also **overfit**.

Overfitting means the model performs well on examples it has already seen but poorly on new examples.

That is why machine learning normally separates data into

training data

and

test data.

The important question is not

“Can the model remember the training flowers?”

but rather

“Can the learned representation help classify flowers the model has never seen before?”

23 A Classroom Experiment

Plot only two Iris features:

$$x = \text{petal length}$$

and

$$y = \text{petal width.}$$

Color each point according to species.

You will often notice that Setosa forms a very clear group, while Versicolor and Virginica overlap more.

Now train a neural network with a two-neuron hidden layer.

Plot the hidden representation

$$(z_1, z_2)$$

instead of the original measurements.

Then ask:

“Are the species more clearly separated in the learned space?”

If they are, you are directly observing representation learning.

24 From Iris Flowers to Modern AI

The Iris example is small.

Modern AI systems may learn representations containing hundreds or thousands of dimensions.

However, the core idea is the same.

For Iris,

4 measurements $\rightarrow$ hidden representation $\rightarrow$ 3 species.

For image recognition,

millions of pixels $\rightarrow$ learned visual representation $\rightarrow$ object category.

For language models,

text $\rightarrow$ learned representations $\rightarrow$ language predictions.

For speech recognition,

audio waveform $\rightarrow$ learned sound representation $\rightarrow$ words.

Representation learning is one of the fundamental ideas behind modern artificial intelligence.

25 The Complete Iris Representation-Learning Pipeline

The whole process can be summarized as:

REAL FLOWER

|
v

Measure the flower

|
v

[sepal length,
sepal width,
petal length,
petal width]

```

      |
      v
x = [5.9, 3.0, 4.2, 1.5]
      |
      v

```

```

NEURAL NETWORK
learned weights
      |
      v

```

```

z = [1.4, -0.8]
learned representation
      |
      v

```

```

class scores
      |
      v

```

```

Setosa      0.05
Versicolor  0.88
Virginica   0.07
      |
      v

```

```

PREDICTION

```

```

Iris versicolor

```

Training determines the weights.

The weights determine the transformation.

The transformation creates the representation.

The representation helps make classification easier.

26 The Most Important Idea

The heart of representation learning can be summarized as

raw data $\rightarrow$ learned transformation $\rightarrow$ useful representation

For the Iris dataset,

[sepal and petal measurements] $\rightarrow$ neural-network weights $\rightarrow$ embedding $\rightarrow$ species

A neural network is not simply memorizing flower names.

It is learning a way to organize information so that differences important for the task become easier to detect.

That is **representation learning**.

27 Final Mental Model

Imagine that the original Iris dataset is a messy desk containing information about flowers.

Representation learning reorganizes the desk.

It does not create new flowers.

It does not magically add information.

Instead, it rearranges and combines the existing information so that important patterns become easier to see.

Remember the process:

Measurements $\rightarrow$ Weights $\rightarrow$ Representation $\rightarrow$ Decision

In plain English:

A neural network learns how to look at the data before it learns how to make the final decision.

That learned way of looking at the data is the **representation**.